Paul Zinn-Justin

Professor, School of Mathematics and Statistics, The University of Melbourne

Curriculum vitae

pzinn@unimelb.edu.au | http://blogs.unimelb.edu.au/paul-zinn-justin/
School of Mathematics and Statistics, The University of Melbourne, VIC 3010, Australia
ORCID | zbMATH | Google Scholar

Professional Experience

2020– Professor at the School of Mathematics and Statistics of the University of Melbourne (Australia).

2016–2020 ARC Future Fellow at the School of Mathematics and Statistics of the University of Melbourne (Australia).

2013–2016 Directeur de recherches (senior scientist) CNRS at LPTHE (UMR 7589), Université Pierre et Marie Curie (France).

2012 "Research Member" of the MSRI program "Random Spatial Processes" (Berkeley).

2009–2010 "Visiting Professor" of the KNAW (Royal Netherlands Academy of Arts and Sciences), visiting the University of Amsterdam.

2008–2013 Chargé de recherches CNRS at LPTHE (UMR 7589), Université Pierre et Marie Curie (France).

2005–2008 Chargé de recherches CNRS at LPTMS (UMR 8626), Université Paris-Sud (France).

2004–2005 Chargé de recherches CNRS at Laboratoire Poncelet (LIFR–MIIP, UMI 2615), Moscow (Russia).

2000–2004 Chargé de recherches (junior scientist) CNRS at LPTMS (UMR 8626), Université Paris-Sud (France).

2000 Visitor at C.N. Yang Institute for Theoretical Physics, SUNY Stony Brook.

1999–2000 Post-doctoral fellow at the New High Energy Theory Center, Rutgers University.

1998–1999 National service.

1995–1998 PhD in theoretical physics at the Laboratoire de Physique Théorique de l'Ecole Normale Supérieure de Paris, supervisor V. Kazakov.

Research Interests

- Algebraic combinatorics, symmetric functions; Schubert calculus, enumerative geometry.
- Exactly solvable models in statistical mechanics, quantum integrability, Yang–Baxter equation, and quantum groups.
- Random matrices, and connections between mathematical physics and enumerative combinatorics.
- Computational and experimental mathematics, including algorithm design, computer algebra and machine learning.

Prizes, Awards

2016–2020	Australian Research Council Future Fellowship "From quantum integrable systems to algebraic geometry and combinatorics".
2011–2016	European Research Council grant "Loop models, Integrability and Combinatorics".
2010–2014	Prime d'excellence scientifique (scientific excellency award) from CNRS.
2009	Visiting Professorship award from the Royal Netherlands Academy of Arts and Sciences.

Teaching Experience

2019–2026	*Discrete Mathematics*, *Linear Algebra*, *Group theory and linear algebra*, *Enumerative combinatorics*, *Exactly solvable models*, *Experimental mathematics* (48h/36h) courses at the University of Melbourne.
2022	LAWRGe lectures at the University of Southern California: *Schubert Calculus and Quantum Integrability* (14h).
2019	Mini-course at the University of Virginia: *Quantum integrability and symmetric polynomials* (10h).
2019	Mini-course at the University of Melbourne: *Schubert calculus* (12h).
2018	Mini-course at the University of Melbourne: *Geometry, Integrability and symmetric functions* (6h).
2015	Lectures at the Department of Mathematics of the Higher School of Economics (Moscow): *Geometry, Integrability and symmetric functions* (8h).
2013–2015	*Intégrabilité quantique*, Masters course at Université Pierre et Marie Curie (30h).
2012	Chern–Simons lectures at the University of California at Berkeley: *Exactly solvable tiling models, Schur functions and the Littlewood–Richardson rule* (6h).
2008	Course at Les Houches summer school: *Integrability and combinatorics: selected topics*.
2007	Lectures at Barcelona University: *Quantum integrability and Razumov–Stroganov type conjectures*.
1997–1998	Teaching Assistant at ENS Paris: mathematics course.
1995–1997	Oral examiner at preparatory school Louis-Le-Grand (bachelor level).

Studies and Academic Honours

2008	Habilitation à Diriger des Recherches "Six-Vertex, Loop and Tiling models: Integrability and Combinatorics" (Université Pierre et Marie Curie, Paris 6).
1998	PhD "Quelques applications de l'Ansatz de Bethe", mention très honorable avec les félicitations du jury (highest mark) (Université Pierre et Marie Curie, Paris 6).
1996	Mathematics "Agrégation" (teaching diploma).
1994–1995	D.E.A. (Master's) in theoretical physics, ENS.
1993–1994	Licence et maîtrise in mathematics, licence in physics (Bachelor's degrees).

1993 Admission at Ecole Normale Supérieure (ENS, Paris) on the Mathematics–Physics–Computer Science entrance exam.

1992 Silver medal at the International Mathematical Olympiads.

Administration of Research

- Organisation of scientific meetings:
  - conference "Formal Power Series and Algebraic Combinatorics 2026", Seattle (Program committee member).
  - conference "At the crossroads of physics and mathematics: the joy of integrable combinatorics — a conference in honour of Philippe Di Francesco's 60th birthday", 24–26 June 2024, Saclay (France).
  - conference "Integrability, Combinatorics and Representations '19", 2–6 September 2019, Giens (France).
  - workshop "Geometric R-matrices", 18–22 December 2017, MATRIX (Australia).
  - program "New trends in integrable models", 15 August–25 November 2016, International Institute of Physics (Natal, Brazil).
  - program "Statistical Mechanics, Integrability and Combinatorics", 11 May–3 July 2015, and conference "Lattice Models: Exact Methods and Combinatorics", 18–22 May 2015, Galileo Galilei Institute (Florence, Italy).
  - conference "Integrability and Combinatorics '14", Giens (France), 23–27 June 2014.
  - conference "Formal Power Series and Algebraic Combinatorics 2013", Paris (Program committee member).
  - conference "Random Matrices and Integrable Systems", Les Houches, 5–9 March 2012.
  - conference "GranMa '11", IHP (Paris), 3–6 October 2011.
  - conference "Geometry and Integrability in Mathematical Physics '08", CIRM Luminy (France), 15–19 September 2008.
  - conference "Geometry and Integrability in Mathematical Physics '06", Moscow (Russia), 15–19 June 2006.
  - workshop "Combinatorial methods in Physics and Knot Theory", Moscow (Russia), 21–25 February 2005.
- Supervision of six PhD students (T. Fonseca 2007–2010, A. Garbali 2012–2015, A. Gunna 2020–2024, A. Stoyan 2023–2026, Junru Hu 2026–, Jakob Sloan 2026–), of ten post-docs (L. Cantini 2006–2008, K. Kalle 2007–2008, M. Wheeler and D. Betea 2012–2014, A. Ponsaing 2013–2014, P. Finch 2014–2015, I. Halacheva 2018–2020, J. Lamers 2020–2021, J.-E. Bourgine 2022–2023, L. Cassia 2023–2025). L. Cantini later received the CNRS Bronze Medal. Twelve of these are still in academia, among whom five have permanent positions; the remaining four successfully transitioned to the private sector. Supervision of seven Master students and three Bachelor students summer internships.
- Recipient of four Discovery Projects "Multivariate polynomials: combinatorics and applications" (2014–2017), "Matrix product multi-variable polynomials from quantum algebras" (2019–2023), "Elliptic Schubert calculus" (2021–2026), "Shuffle algebras and vertex models" (2024–2026), and a Future Fellowship "From quantum integrable systems to algebraic geometry and combinatorics" (2016–2020) from the ARC.
- Recipient of an ERC consolidator grant for the project "Loop models, Integrability and Combinatorics", 2011–2016 (budget: 900k€).
- "Scientist in charge" of the Marie Curie Research Training Network "ENRAGE" (Random Geometry and Random Matrices: From Quantum Gravity to Econophysics), 2005–2009.

- Coordinator of the ANR program "GIMP" (Geometry and Integrability in Mathematical Physics), 2006–2009.
- Former editor for *Communications in Mathematical Physics*; editor for *Algebraic Combinatorics*.
- Referee for numerous journals (Commun. Math. Phys., Lett. Math. Phys., Electronic J. of Combinatorics, Adv. Appl. Math., J. of Combin. Theory A, JSTAT, J. of Physics A, etc); expert for several research organisations (ANR – France, NWO – Netherlands, ESF – Europe, ARC – Australia).
- Member of the Committee of Academic Sponsors of SLMath (Berkeley). Former member of the Diversity and Inclusion committee (Melbourne University). Former member of the laboratory council (LPTMS and LPTHE).
- PhD referee (A. Ponsaing, L. Poulain d'Andecy).
- Member of the European Marie Curie Research Training Network "ENIGMA", 2005–2008, of the ESF program "MISGAM", 2005–2009, and of the ANR program "GranMa", 2009–2011.

Summary of Publications

- 80 publications in major international peer-reviewed journals.
- 7 chapters of books.
- 8 preprints.
- 1 popular science article.
- "Habilitation à Diriger des Recherches" thesis was published as a monograph.
- h-index: 33 (Google Scholar, April 2026)

Summary of Conferences

- Organised conferences: 12
- Invited oral presentations in international conferences and schools: 98
- Other oral presentations: 67

Selected Invited and Plenary Conference Talks

- Conference "FPSAC", UC Davis (USA), 17–21 July 2023: *Schubert puzzles as exactly solvable models* (plenary talk).
- International Congress of Mathematical Physics, Montreal (Canada), 23–28 July 2018: *From quantum integrability to Schubert calculus*.
- Conference "Integrability and Algebraic Combinatorics", IPAM UCLA (USA), 15–19 April 2024: *The MacNeille completion of the type B Bruhat poset*.
- Conference "Quantum integrability and quantum Schubert calculus", Kavli Royal Society Centre (UK), 11–13 June 2018: *Schubert puzzles and quantum integrability*.
- Symposium FoPM Tokyo (Japan), 6–8 February 2023: *From exactly solvable models of statistical mechanics to combinatorics*.
- Main speaker at the 76th "Séminaire Lotharingien de Combinatoire", 3–6 April 2016: *Razumov–Stroganov Correspondences and the Geometry of Schubert Varieties*.

Programming, Web Presence

- Professional expertise in algorithm design.
- High level programming skills in:
  - Traditional programming languages: python, c, c++, C#, Lisp, assembly language. . . ;
  - Web-based languages and related: HTML/CSS/JavaScript/TypeScript, PHP/SQL;
  - Mathematics-related languages: Macaulay2, Mathematica.
- Contributes to various open source projects.
  Developer for (since 2025, "author of") the open source software "Macaulay2":
  - Bug fixes, implementation of new functions.
  - Rewrite of the output engine.
  - Macaulay2Web, a web-based interface using the KaTeX technology.
- Contributes entries to the On-Line Encyclopedia of Integer Sequences.
- Various projects can be found on my website and on my GitHub page.